CLOUD SECURITY — PRACTICAL-BASED LEARNING

PBL 3: Configure a Cloud-Based Attacker Lab and Test Reverse-Shell Connectivity

Standalone practical report

1. Aim

Create an isolated cloud-based security-testing environment using an EC2 instance, restrict SSH access to an authorized public IP, and test TCP connectivity to a controlled, authorized system.

2. Problem Statement

Configure a Linux EC2 instance as a security-testing machine, use an explicit SSH IP allowlist, deploy a Netcat listener for a harmless test, and document connection results and reverse-shell-related security risks.

3. Tools and Technologies

- AWS EC2 and Security Groups
- Amazon Linux
- SSH and an EC2 key pair
- Netcat
- Windows PowerShell
- TCP/IP networking

4. Procedure

Step 1 — Launch Linux EC2

Launch a Linux instance, configure its key pair, and record its current public IPv4 address.

Step 2 — Restrict SSH

Configure inbound TCP 22 so the source is the authorized client's current public IP address using /32 notation.

Step 3 — Connect over SSH

Use the matching key pair, correct Linux username and current EC2 public IP.

Step 4 — Configure a Netcat listener

Configure a listener on the selected TCP port in the isolated lab environment. Use only systems you own or are explicitly authorized to test.

Step 5 — Test TCP connectivity

Use PowerShell to test the target address and port. The initial test returned TcpTestSucceeded : False.

Step 6 — Send a harmless message

The troubleshooting history records that the message PBL3 connectivity test was received by the Netcat listener after troubleshooting.

5. Errors Encountered and Solutions

Incorrect private IP in SSH allowlist

The SSH source was initially configured as 192.168.0.101/32. This is a private LAN address and is generally not the source IP that AWS sees for an internet-facing connection. Correct the rule to the authorized client's current public IP/32.

TCP connectivity failure

PowerShell reported TcpTestSucceeded : False. Check destination IP and port, listener state, instance state, security-group rules and the intended network path. Retry only after confirming the listener is active and the traffic is authorized.

Public versus private address confusion

Use the client’s public IP for an internet-facing SSH allowlist. Use the correct destination address for the chosen path and do not assume a private address is reachable over the public internet.

TCP test versus reverse shell

Receiving plain text over TCP shows that data was transmitted to a listener. It does not prove that an interactive reverse shell was established. The available history contains no verified evidence of an interactive command session.

6. Result

The SSH allowlist error was identified and corrected, and the initial TCP connectivity failure was documented. The recorded message PBL3 connectivity test was subsequently received by the Netcat listener.

VERIFIED SCOPE: The recorded message supports TCP communication. UNVERIFIED: Successful interactive reverse-shell execution. Do not claim a reverse shell unless separate authorized evidence demonstrates an interactive session.

7. Security Measures

- Restrict SSH to the authorized public IP/32.
- Use only disposable, owned or explicitly authorized test systems.
- Limit listener ports to the required lab traffic.
- Protect SSH private keys and credentials.
- Remove temporary listener rules and resources after testing.
- Do not execute reverse-shell payloads on shared, institutional or production systems.

8. Conclusion

The practical documents a cloud-based testing environment, SSH source-IP restrictions and TCP connectivity troubleshooting. The incorrect private-IP allowlist and initial TCP failure were identified. The harmless message provides evidence of TCP communication only; it does not establish successful interactive reverse-shell execution.

9. Screenshots Checklist

- ■ EC2 instance state and public IP.

- Initial incorrect SSH source rule, if available.
- Corrected SSH rule restricted to public IP/32.
- SSH terminal session, if available.
- PowerShell output showing TcpTestSucceeded : False.
- Netcat listener configuration.
- Listener output showing PBL3 connectivity test.
- Security-group rules and cleanup evidence.